

Hybrid Neighborhood Evaluation with focus on the Fibonacci-distance Neighborhood for the Permutation Flow Shop Problem

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Introduction

- **Permutation Flow Shop Problem (PFSP):** NP-hard for $m \geq 3$. Neighborhood choice drives metaheuristic quality.
- Classical neighborhoods trade off coverage and cost:
 - **Adjacent swap:** $n-1$ moves, only short-range.
 - **Dynasearch:** $O(n^2)$ candidates, expensive per call.
 - **Motzkin arcs:** structured but $O(n^3)$ DP — breaks beyond $n \sim 100$.
- **Fibonacci-distance insertion:** $O(n \log n)$ moves spanning short *and* long range. Stays tractable up to $n = 500$ and beyond.
- This talk: define the Fibonacci neighborhood, position it among the other elementary moves, and show its role within hybrid local search (TS, SA).

The Fibonacci-distance Neighborhood

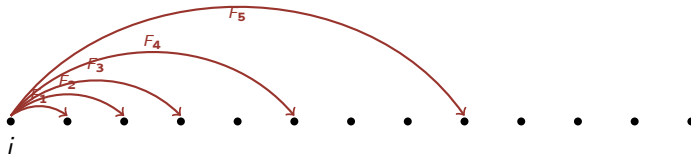
Definition. Given $\pi = (\pi_0, \dots, \pi_{n-1})$, define moves

$$N_{\text{fib}}(\pi) = \{ \text{insert}(\pi_i \rightarrow i + F_k) : 0 \leq i < n, F_k \leq n - i \},$$

where $F_k = 1, 2, 3, 5, 8, 13, 21, \dots$ are Fibonacci numbers.

Size: $|N_{\text{fib}}(\pi)| = O(n \log n)$. Local + long-range coverage from a single rule.

Idea. A geometric, golden-ratio-scaled set of move distances from position i :



Why Fibonacci? — Properties

- **Golden-ratio scaling.** Consecutive jump distances grow by $\varphi \approx 1.618$ — the densest non-uniform integer sequence avoiding any short cycle.
- **Logarithmic per-position.** Only $O(\log n)$ jump lengths per starting position; total $O(n \log n)$ moves vs. $O(n^2)$ for full insertion or dynasearch.
- **Tridiagonal QUBO.** The QUBO matrix has bandwidth 2 — embeds natively on D-Wave with no chain breaks. Scales cleanly to $n = 500$.
- **Self-similarity.** Fibonacci shifts within a Fibonacci shift form a sub-neighborhood — the move structure is recursive, well-suited to multi-scale local search.

Role in Hybrid Strategies

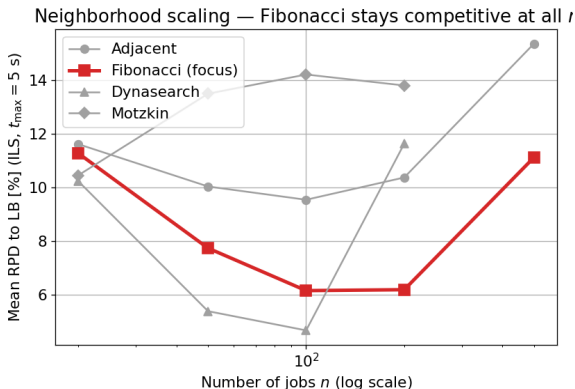
Four elementary neighborhoods are combined under five strategies (FC, PW, PA, BK, DP). Fibonacci's job in each:

- **FC — Fixed Cycle:** Fibonacci breaks adjacent-swap stagnation between two dynasearch passes.
- **PA — Performance-Adaptive:** the EMA-driven selector picks Fibonacci most often in mid-search, once adjacent saturates but before dynasearch becomes affordable.
- **BK — Best-of- K :** Fibonacci is the cheapest non-trivial neighborhood to evaluate, so its candidates effectively come for free in the BK sweep.
- **Large n :** for $n \geq 200$ Fibonacci is the only elementary neighborhood that completes *multiple* iterations within the time budget — the rest degenerate to a single call.

Experimental Setup

- **Benchmarks:** Taillard instances, $n \in \{20, 50, 100, 200, 500\}$, $m \in \{5, 10, 20\}$, 10 instances each, 5 seeds.
- **Time limits:** $t_{\max} \in \{0.1, 0.5, 1, 2, 5, 10\}$ s; metric: mean RPD to lower bound.
- **Algorithms:** Iterated Local Search (tabu $\tau=10$, MushroomList $k=10$); Simulated Annealing ($T_0=1000$, $\alpha=0.99$, reheat-kick on stagnation).
- **Quantum:** D-Wave Advantage 4.1, 5,000 reads, windowed decomposition for $n > 19$.
- **Compared neighborhoods:** Adjacent, **Fibonacci**, Dynasearch, Motzkin.

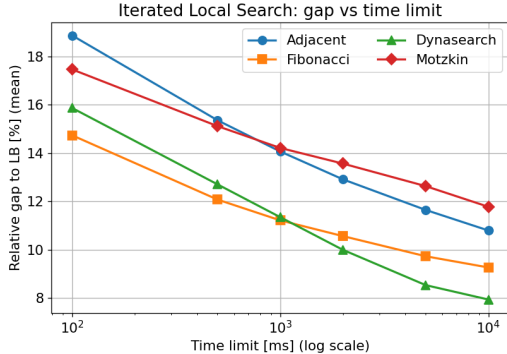
Scaling Behavior — RPD vs. n



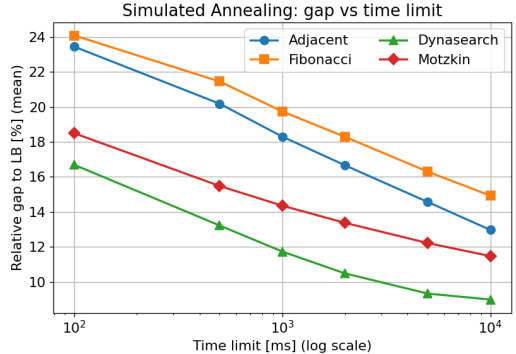
Observations (ILS, $t_{\max} = 5$ s, $m=10$):

- Up to $n \approx 100$: Dynasearch wins, Fibonacci second.
- $n = 200$: Fibonacci **overtakes** Dynasearch (6.19 % vs. 11.63 %).
- $n = 500$: Dynasearch and Motzkin become intractable; Fibonacci continues at **11.12 %** — the only classical neighborhood that still scales.

Gap vs. Time Limit — Fibonacci within the Hybrid Pool



ILS



SA

- Fibonacci converges **fastest** (flat curve from $t_{\max} = 100$ ms onward) but plateaus highest — ideal as the *warm-up* neighborhood within a hybrid

Conclusion and Future Work

Summary:

- Fibonacci is the only elementary neighborhood that stays tractable from $n = 20$ to $n = 500$.
- Tridiagonal QUBO structure makes it the natural classical baseline for D-Wave QPU comparison.
- Inside a hybrid strategy (PA, BK), Fibonacci absorbs the early time budget cheaply; heavier neighborhoods then refine.
- Empirically, Fibonacci wins outright above $n = 200$ where its rivals run out of computational budget.

Future work:

- Quantum-assisted Fibonacci neighborhood (windowed QUBO, n up to 500).
- Multi-armed bandit selection over neighborhoods with Fibonacci as the low-cost arm.